

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

1) Which of the following statements concerning the breakdown of glucose to CO_2 and water is (are) *true*?

- A) Adding electrons to another substance is known as reduction.
- B) The breakdown of glucose is exergonic.
- C) An electron acceptor is called the reducing agent.
- D) A and B only are correct.
- E) A, B, and C are correct.

2) The oxygen consumed during cellular respiration is directly involved in

- A) the oxidation of pyruvate to acetyl CoA.
- B) accepting electrons at the end of the electron transport chain.
- C) glycolysis.
- D) the citric acid cycle.
- E) the phosphorylation of ADP.

3) All of the following statements about NAD^+ are true *except*:

- A) NAD^+ is reduced to NADH during both glycolysis and the Krebs cycle.
- B) NAD^+ is reduced by the action of dehydrogenases.
- C) NAD^+ can receive electrons for use in oxidative phosphorylation.
- D) NAD^+ has more chemical energy than NADH.
- E) In the absence of NAD^+ , glycolysis cannot function.

4) Glycolysis is believed to be one of the most ancient of metabolic processes. Which statement below *least* supports this idea?

- A) Glycolysis is found in all eukaryotic cells.
- B) Bacteria, the most primitive of cells, make extensive use of glycolysis.
- C) The enzymes of glycolysis are found in the cytosol rather than in a membrane-enclosed organelle.
- D) Glycolysis neither uses nor needs O_2 .
- E) If run in reverse, glycolysis will build glucose molecules.

5) Which kind of metabolic poison would most directly interfere with glycolysis?

- A) an agent that binds to pyruvate and inactivates it
- B) an agent that reacts with oxygen and depletes its concentration in the cell
- C) an agent that closely mimics the structure of glucose but is not metabolized
- D) an agent that inhibits the formation of acetyl coenzyme A
- E) an agent that reacts with NADH and oxidizes it to NAD^+

6) Which process in eukaryotic cells will normally proceed whether O_2 is present or absent?

- A) oxidative phosphorylation
- B) fermentation
- C) electron transport
- D) glycolysis
- E) the Krebs cycle

7) All of the following statements about glycolysis are true *except*:

- A) The enzymes of glycolysis are located in the cytosol of the cell.
- B) Glycolysis can operate in the complete absence of O_2 .
- C) Glycolysis has steps involving oxidation-reduction reactions.
- D) The end products of glycolysis are CO_2 and H_2O .
- E) Glycolysis makes ATP exclusively through substrate-level phosphorylation.

8) All of the following substances are produced in a muscle cell under anaerobic conditions *except*

- A) NADH.
- B) lactate.
- C) acetyl CoA.
- D) pyruvate.
- E) ATP.

9) In addition to ATP, what are the end products of glycolysis?

- A) CO₂ and H₂O
- B) H₂O and ethyl alcohol
- C) NADH and pyruvate
- D) CO₂ and ethyl alcohol
- E) CO₂ and NADH

10) All of the following are products or intermediaries in glycolysis *except*

- A) pyruvate.
- B) phosphoenolpyruvate.
- C) NADH.
- D) ATP.
- E) FADH₂.

11) All of the following are functions of the Krebs cycle *except*

- A) production of ATP.
- B) adding electrons and protons to oxygen to form water.
- C) production of FADH₂.
- D) production of NADH.
- E) release of carbon dioxide.

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.
Refer to Figure 9.1 to answer the following questions. In Figure 9.1 there are some reactions of glycolysis in their proper sequence. Each reaction is lettered. Use these letters to answer the questions.



Figure 9.1

12) Which reaction shows a split of one molecule into two smaller molecules?

13) In which reaction is an inorganic phosphate added to the reactant?

- 14) In which reaction is a net gain of ATP finally realized from glycolysis?
- 15) Which step requires ATP to occur?
- 16) Which step is the first to produce an acid?

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17) The Krebs cycle produces which of the following molecules that then transfer energy to the electron transport system?

- A) NADH and ATP
 B) CO_2 and FAD
 C) FADH_2 and NADH
 D) ATP and CO_2
 E) NADH, FADH_2 , and ATP

18) Which of the following intermediary metabolites enters the Krebs cycle and is formed, in part, by the removal of CO_2 from a molecule of pyruvate?

- A) glyceraldehyde phosphate
 B) oxaloacetic acid
 C) acetyl CoA
 D) lactate
 E) citric acid

19) A young relative of yours has never had much energy. He goes to a doctor for help and is sent to the hospital for some tests. There they discover his mitochondria can use only fatty acids and amino acids for respiration, and his cells produce more lactate than normal. Of the following, which is the best explanation of his condition?

- A) His cells contain something that inhibits oxygen use in his mitochondria.
 B) His cells cannot move NADH from glycolysis into the mitochondria.
 C) His cells lack the enzyme in glycolysis that forms pyruvate.
 D) His mitochondria lack the transport protein that moves pyruvate across the outer mitochondrial membrane.
 E) His cells have a defective electron transport chain, so glucose goes to lactate instead of to acetyl CoA.

20) Each time a molecule of glucose is completely oxidized via aerobic respiration, how many oxygen (O_2) molecules are required?

- A) 2 B) 6 C) 38 D) 1 E) 12

Refer to Figure 9.2 as a guide to answer the following questions.

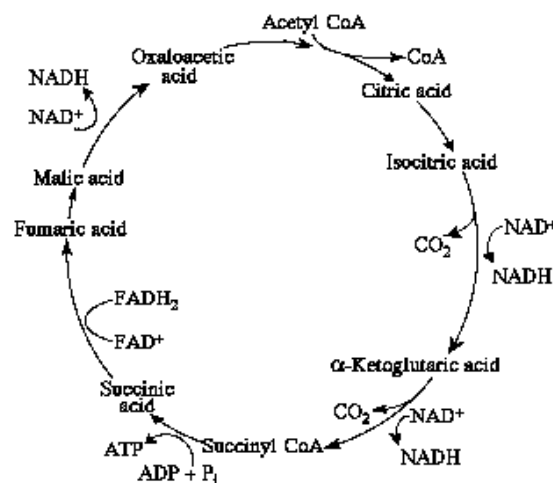


Figure 9.2

21) Starting with fumaric acid and ending with α -ketoglutaric acid in Figure 9.2, what is the maximum number of ATP molecules that could be made through substrate-level phosphorylation?

- A) 2
- B) 1**
- C) 12
- D) 11
- E) more than 20

22) Refer to Figure 9.2. Carbon skeletons for amino acid biosynthesis are supplied by intermediates of the Krebs cycle. Which intermediate would supply the carbon skeleton for synthesis of a 5-carbon amino acid?

- A) malic acid
- B) citric acid
- C) α -ketoglutaric acid**
- D) succinic acid
- E) isocitric acid

23) Refer to Figure 9.2. Starting with citric acid and ending with oxaloacetic acid, how many ATP molecules can be formed from oxidative phosphorylation?

- A) 38 B) 1 C) 15 **D) 11** E) 5

24) Refer to Figure 9.2. Where do the catabolism products of fatty acid breakdown enter into the Krebs cycle?

- A) α -ketoglutaric acid
- B) acetyl CoA
- C) pyruvate**
- D) succinyl CoA
- E) glyceraldehyde phosphate

25) Refer to Figure 9.2. How many ATP molecules can be made through substrate-level phosphorylation and oxidative phosphorylation if you started with succinyl CoA and ended with oxaloacetate?

- A) 6** B) 5 C) 9 D) 7 E) 15

26) During oxidative phosphorylation, H_2O is formed. Where do the oxygen atoms in the H_2O come from?

- A) lactate
- B) molecular oxygen**
- C) carbon dioxide
- D) glucose
- E) pyruvate

27) Cellular respiration harvests the most chemical energy from which process?

- A) converting oxygen to ATP
- B) generating carbon dioxide and oxygen in the electron transport chain
- C) transferring electrons from organic molecules to oxygen**
- D) substrate-level phosphorylation
- E) forming lactate from pyruvate

28) During aerobic respiration, electrons travel downhill in which sequence?

- A) glucose $\rightarrow$ ATP $\rightarrow$ oxygen
- B) glucose $\rightarrow$ ATP $\rightarrow$ electron transport chain $\rightarrow$ NADH
- C) food $\rightarrow$ NADH $\rightarrow$ electron transport chain $\rightarrow$ oxygen**
- D) food $\rightarrow$ Krebs cycle $\rightarrow$ ATP $\rightarrow$ NAD $^+$
- E) food $\rightarrow$ glycolysis $\rightarrow$ Krebs cycle $\rightarrow$ NADH $\rightarrow$ ATP

29) During aerobic respiration, which of the following directly donates electrons to the electron transport chain at the lowest energy level?

A) NAD^+ B) ATP C) FADH_2 D) $\text{ADP} + \text{P}_i$ E) NADH

30) Which metabolic process is most closely associated with intracellular membranes?

- A) substrate-level phosphorylation
- B) oxidative phosphorylation
- C) ethanolic fermentation
- D) glycolysis
- E) the Krebs cycle

31) In chemiosmotic phosphorylation, what is the most direct source of energy that is used to convert $\text{ADP} + \text{P}_i$ to ATP?

- A) energy released from substrate-level phosphorylation
- B) energy released as electrons flow through the electron transport system
- C) energy released from movement of protons through ATP synthase
- D) energy released from ATP synthase pumping hydrogen ions against their concentration gradient
- E) No external source of energy is required because the reaction is exergonic.

32) Hydrogen ions flowing through a cylinder causing the cylinder and attached rod to rotate like rushing steam turns a water wheel is an analogy that best describes

- A) formation of NADH in glycolysis.
- B) the Krebs cycle.
- C) ATP synthase activity.
- D) the electron transport system.
- E) oxidative phosphorylation.

33) During aerobic cellular respiration, a proton gradient in mitochondria will be generated by _____ and used primarily for _____.

- A) the electron transport chain ... ATP synthesis
- B) fermentation ... NAD reduction
- C) the electron transport chain ... substrate-level phosphorylation
- D) diffusion of protons ... ATP synthesis
- E) glycolysis ... production of H_2O

34) The direct energy source that drives ATP synthesis during respiratory oxidative phosphorylation is

- A) the thermodynamically favorable flow of electrons from NADH to the mitochondrial electron transport carriers.
- B) the difference in H^+ concentrations on opposite sides of the inner mitochondrial membrane.
- C) the final transfer of electrons to oxygen.
- D) oxidation of glucose to CO_2 and water.
- E) the thermodynamically favorable transfer of phosphate from glycolysis and the Krebs cycle intermediate molecules of ADP.

35) A major function of the mitochondrial inner membrane is the conversion of energy from electrons to the stored energy of the phosphate bond in ATP. To accomplish this function, this membrane must have all of the following features *except*

- A) proton pumps embedded in the membrane.
- B) integral, transverse ATP synthase.
- C) the electron transport chain of proteins.
- D) proteins to accept electrons from NADH.
- E) high permeability to protons.

36) When hydrogen ions are pumped from the mitochondrial matrix across the inner membrane and into the intermembrane space, the result is

- A) the formation of ATP.
- B) the creation of a proton gradient.
- C) the restoration of the $\text{Na}^+ - \text{K}^+$ balance across the membrane.

- D) the reduction of NAD^+ .
- E) the lowering of pH in the mitochondrial matrix.

37) Where is ATP synthase located in the mitochondrion?

- A) cytochrome system
- B) inner membrane
- C) ribosomes
- D) outer membrane
- E) matrix

38) The primary role of oxygen in respiration is to

- A) act as an acceptor for electrons and hydrogen, forming water.
- B) catalyze the glycolysis reaction.
- C) combine with carbon, forming CO_2 .
- D) combine with lactic acid to form pyruvic acid.
- E) yield energy in the form of ATP as it is passed down the respiratory chain.

39) It is possible to prepare vesicles from portions of the inner membrane of the mitochondrial components. Which one of the following processes could still be carried on by this isolated inner membrane?

- A) the Krebs cycle
- B) glycolysis and fermentation
- C) oxidative phosphorylation
- D) both the Krebs cycle and oxidative phosphorylation
- E) reduction of NAD

40) The primary function of the mitochondrion is the production of ATP. To carry out this function, the mitochondrion must have all of the following *except*

- A) the membrane-bound electron transport chain.
- B) enzymes for glycolysis.
- C) enzymes for the Krebs cycle.
- D) proton pumps embedded in the inner membrane.
- E) mitochondrial ATP synthase.

41) How many moles of ATP are produced from the complete oxidation of a mole of glucose in cellular respiration?

- A) 20 B) 15 C) 12 D) 30 E) 36

42) How many carbon atoms feed into the Krebs cycle?

- A) 4 B) 10 C) 6 D) 2 E) 8

43) Inside an active mitochondrion, most electrons follow which pathway?

- A) electron transport chain $\rightarrow$ Krebs cycle $\rightarrow$ ATP
- B) glycolysis $\rightarrow$ NADH $\rightarrow$ oxidative phosphorylation $\rightarrow$ ATP $\rightarrow$ oxygen
- C) pyruvate $\rightarrow$ Krebs cycle $\rightarrow$ ATP $\rightarrow$ NADH $\rightarrow$ oxygen
- D) Krebs cycle $\rightarrow$ FADH_2 $\rightarrow$ electron transport chain $\rightarrow$ ATP
- E) Krebs cycle $\rightarrow$ NADH $\rightarrow$ electron transport chain $\rightarrow$ oxygen

44) How many molecules of CO_2 would be released from the complete aerobic respiration of a molecule of sucrose, a disaccharide?

- A) 2 B) 6 C) 12 D) 3 E) 38

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

The questions below are based on the stages of glucose oxidation listed below.

A. stage I: glycolysis

- B. stage II: oxidation of pyruvate to acetyl CoA
- C. stage III: Krebs cycle
- D. stage IV: oxidative phosphorylation (chemiosmosis)

45) Which one of the stages produces the most ATP when glucose is completely oxidized to carbon dioxide and water?

46) Which one of the stages normally occurs whether or not oxygen is present?

47) Which one of the stages occurs in the cytosol of the cell?

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

48) Carbon dioxide is released during which stage(s)?

- A) stages III and IV
- B) stages II and III
- C) stages I, II, and III
- D) stage III only
- E) stages II, III, and IV

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The questions below will use the following key:

- A. cytosol
- B. mitochondrial outer membrane
- C. mitochondrial inner membrane
- D. mitochondrial intermembrane space
- E. mitochondrial matrix

49) The electron transport chain energy is used to pump H^+ ions into which location?

50) Glycolysis takes place in which location?

51) Where are the proteins of the electron transport chain located?

52) Electron shuttle across the membrane begins in which location?

53) Acetyl CoA accumulates in which location?

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54) Which of the following statements is true of fermentation?

- A) It produces more energy per glucose molecule than does aerobic respiration.
- B) It can be performed only by bacteria.
- C) It produces a net gain of ATP.
- D) It is an aerobic process.
- E) It produces a net gain of NADH.

55) Muscle cells in oxygen deprivation convert pyruvate to _____ and in this step gain _____.

- A) lactate ... NAD^+
- B) alcohol ... CO_2
- C) lactate ... ATP
- D) ATP ... NAD^+
- E) alcohol ... ATP

56) The ATP made during fermentation is generated by which of the following?

- A) electron transport
- B) oxidation of NADH
- C) photophosphorylation
- D) substrate-level phosphorylation
- E) chemiosmosis

57) What must a muscle cell do to keep contracting?

- A) rapidly replace the enzymes of respiration
- B) effectively transport NADH into the mitochondria
- C) store sufficient ATP as a raw material for oxidative phosphorylation
- D) regenerate ATP at a very fast rate
- E) receive sufficient oxygen for glycolysis

58) Which of the following is *not* true concerning the cellular compartmentation of the steps of respiration or fermentation?

- A) NADH is produced only in the mitochondria.
- B) Lactate is produced only in the cytosol.
- C) Acetyl CoA is produced only in the mitochondria.
- D) FADH₂ is produced only in the mitochondria.
- E) ATP is produced in the cytosol and the mitochondria.

59) An organism is discovered that consumes a considerable amount of sugar, yet does not gain much weight when denied air. Curiously, the consumption of sugar increases as air is removed from the organism's environment, but the organism seems to thrive even in the absence of air. When returned to normal air, the organism does fine. Which of the following best describes the organism?

- A) It must use a molecule other than oxygen to accept electrons from the electron transport chain.
- B) It is a normal eukaryotic organism.
- C) The organism obviously lacks the Krebs cycle and electron transport chain.
- D) It is a facultative anaerobe.
- E) It is an anaerobic organism.

60) Catabolism of proteins, lipids, and carbohydrates can result in a 2-carbon molecule that enters the Krebs cycle. What is the molecule?

- A) glucose
- B) acetic acid or acetyl CoA
- C) an amino acid
- D) pyruvate
- E) a fatty acid

61) Carbon skeletons to be broken down during cellular respiration can be obtained from

- A) proteins.
- B) polysaccharides.
- C) lipids.
- D) A and B.
- E) A, B, and C.

62) Which of the following does *not* participate in glycolysis?

- A) fatty acids
- B) sucrose
- C) glycerol
- D) starch
- E) glucose

63) You have a friend who lost 15 pounds of fat on a diet. Where did the fat go (how was it lost)?

- A) It was converted to urine and eliminated from the body.
- B) Chemical energy was converted to heat and then released.
- C) It was broken down to amino acids and eliminated from the body.
- D) It was released as CO₂ and H₂O.

E) It was converted to ATP, which weighs much less than fat.

64) Phosphofructokinase is an important control enzyme. All of the following statements concerning this enzyme are true *except*:

A) It is activated by citrate.

B) It is inhibited by ATP.

C) It is an allosteric enzyme.

D) It is a coordinator of the processes of glycolysis and the Krebs cycle.

E) It is activated by ADP.

- 1) Answer: D
- 2) Answer: B
- 3) Answer: D
- 4) Answer: E
- 5) Answer: C
- 6) Answer: D
- 7) Answer: D
- 8) Answer: C
- 9) Answer: C
- 10) Answer: E
- 11) Answer: B
- 12) Answer: B
- 13) Answer: C
- 14) Answer: E
- 15) Answer: A
- 16) Answer: D
- 17) Answer: C
- 18) Answer: C
- 19) Answer: D
- 20) Answer: B
- 21) Answer: B
- 22) Answer: C
- 23) Answer: D
- 24) Answer: C
- 25) Answer: A
- 26) Answer: B
- 27) Answer: C
- 28) Answer: C

- 29) Answer: C
- 30) Answer: B
- 31) Answer: C
- 32) Answer: C
- 33) Answer: A
- 34) Answer: B
- 35) Answer: E
- 36) Answer: B
- 37) Answer: B
- 38) Answer: A
- 39) Answer: C
- 40) Answer: B
- 41) Answer: E
- 42) Answer: D
- 43) Answer: E
- 44) Answer: C
- 45) Answer: D
- 46) Answer: A
- 47) Answer: A
- 48) Answer: B
- 49) Answer: D
- 50) Answer: A
- 51) Answer: C
- 52) Answer: B
- 53) Answer: E
- 54) Answer: C
- 55) Answer: A
- 56) Answer: D

57) Answer: D

58) Answer: A

59) Answer: D

60) Answer: B

61) Answer: E

62) Answer: A

63) Answer: D

64) Answer: A